

BioTDR & Immunology Development Group for Engagement

BRIDGE 2021 Q2 Seminar: Big Picture Drug Discovery

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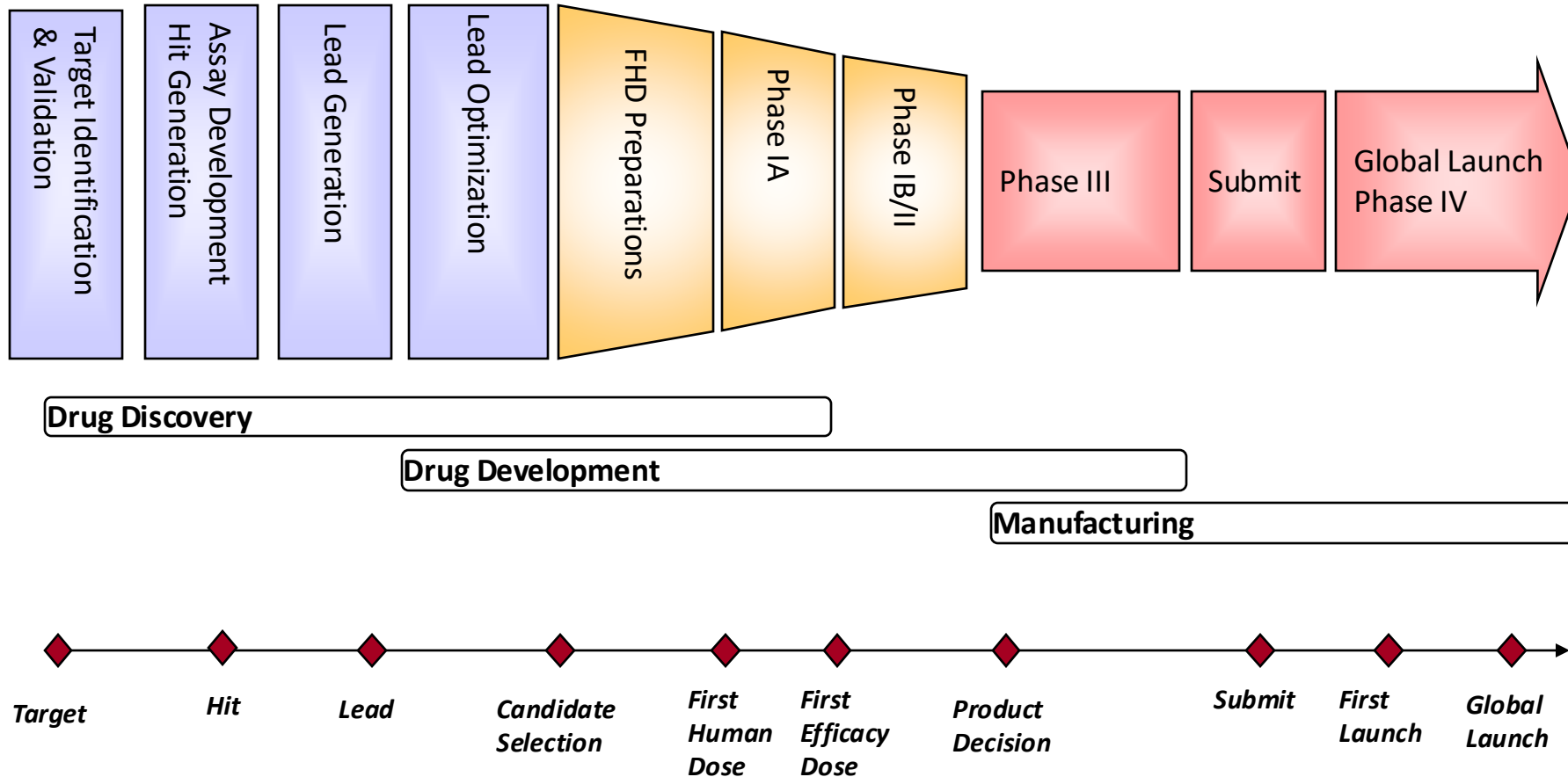
Lilly

Drug Discovery and Development Research Overview

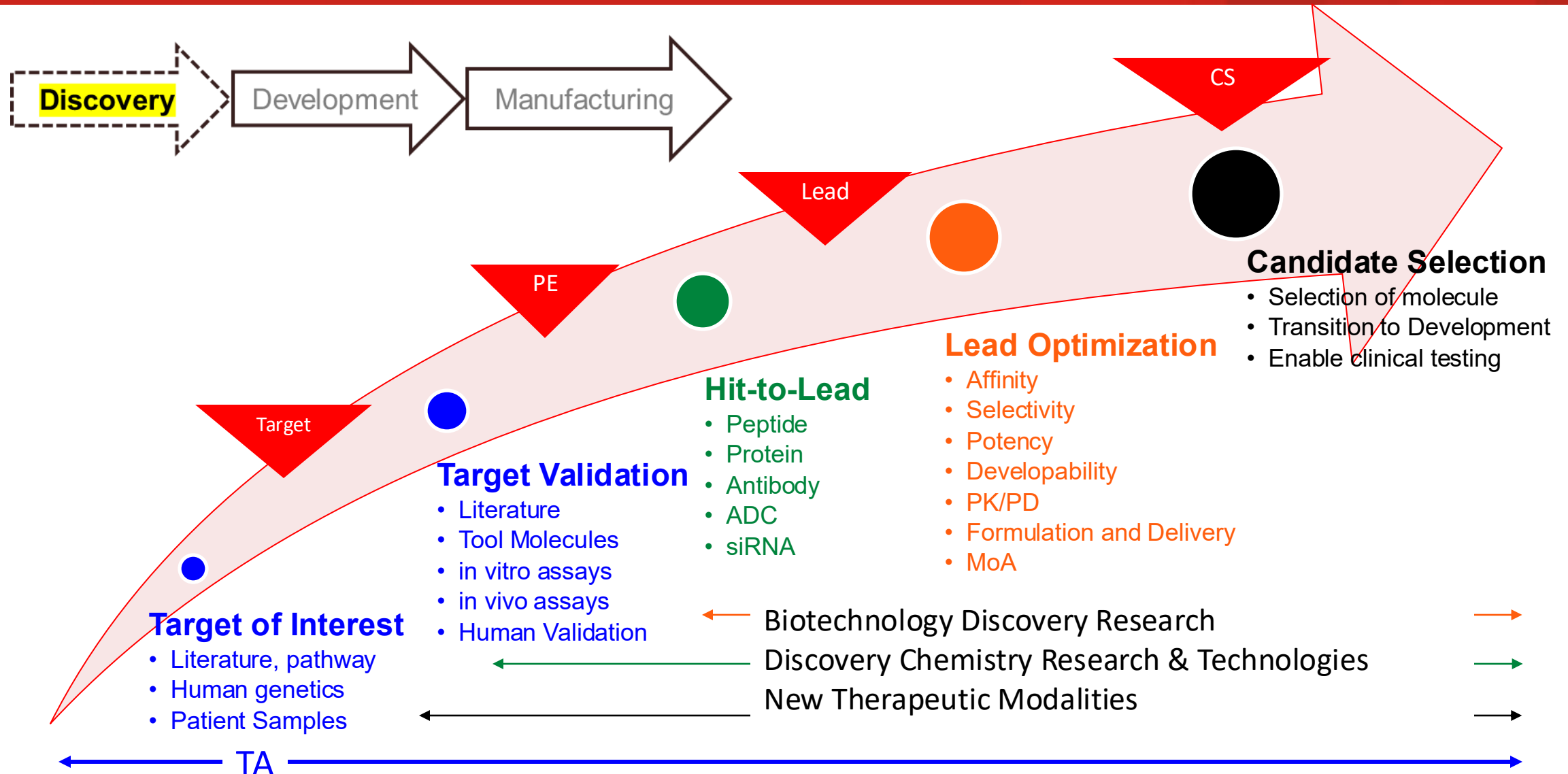
Hypothesis Generation

Candidate Development

Commercialisation

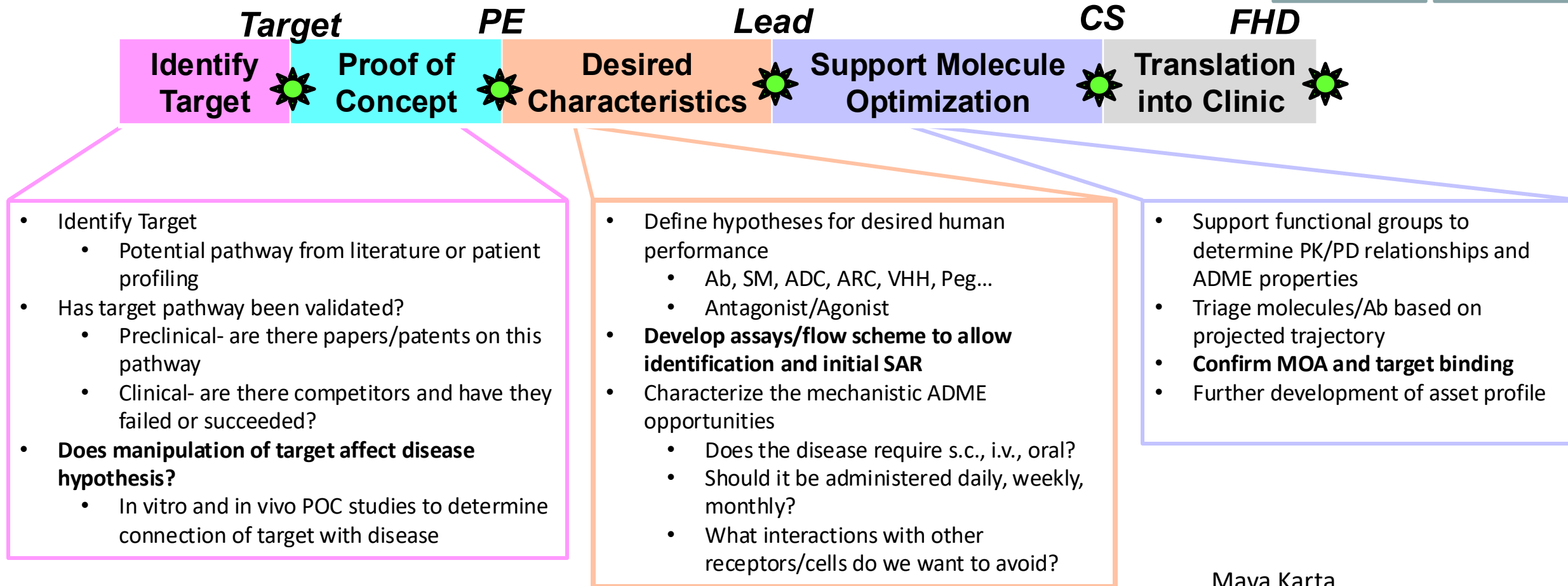
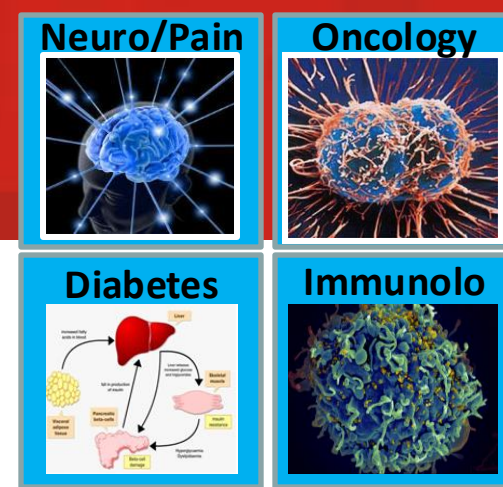


Discovery Process: Target to Candidate Selection



TA's role in drug discovery (ex. Immunology)

- Propose and validate targets of interest
- Provide clear molecule goals to inform engineers and chemists of desired characteristics needed for optimal molecule design
- Proof of Concept experiments both for disease relevance and MOA



Translational Science Requires Teamwork

Target

PE

Lead

CS

FHD

Identify Target



Proof of Concept



Desired Characteristics



Support Molecule Optimization



Translation into Clinic



- Integration of Discovery and Clinical translational research through Informatics and Technology

• Program teams

• QC
• Enablement

Technical expertise

Logistical support and coordination

Internal data/samples

• Target validation
• New target ID
• Data bases

Bio-informatics

External collaborations

• Disease understanding

Discovery
Clin Pharm
BU

Immunology-Translational WG

Statistics
CLO
LEM
Clinical

Implementation of plan in CT

• Preclinical

- Development and implementation of Target Engagement (TE) and PD assays
- Interface with Clinical Research Organizations (CRO) to transfer and validate assays
- Helps decision making in early phase clinical trials

• Phase I/II

- Execute Mechanism of Action (MOA)-linked data generation alongside TE and PD assays

• Phase III

- Deliver focused analysis to inform Business Unit on LY's MOA
- Support discovery efforts to ID new targets and learn more about disease process

MIH (Medicines Innovation Hub)

- ◆ DCRT (Discovery Chemistry Research Technologies)
- ◆ **BioTDR (BioTechnology Discovery and Research)**
- ◆ NTM (New Therapeutic Modalities)
- ◆ BR&D (BioProduct Research and Development)
- ◆ SMDD (Small Molecule Design and Development)
- ◆ DDCS (Delivery, Device, and Connected Solutions)
- ◆ PD (Product Delivery)

To deliver in the following Therapeutic Areas:
Immunology, Oncology (Loxo), Neurodegeneration, Diabetes & Pain

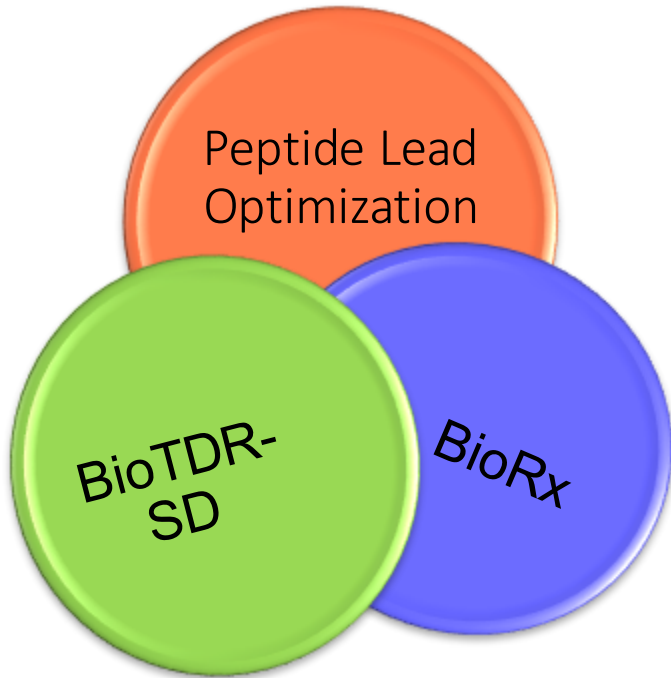
+ Supporting Groups

ATPLV

- ADME (Absorption, Distribution, Metabolism, Excretion)
- Toxicology
- PK/PD (Pharmacokinetics/ Pharmacodynamics)
- LEM (Laboratory for Experimental Medicine)
- Veterinary Resources

CSSP (Clinical Systems & Supply Planning)

BioTDR is a Team of Teams:



Lilly Biotechnology Center (LBC or AME) – San Diego

- Discovery & Automation
- Protein Engineering
- Protein Characterization

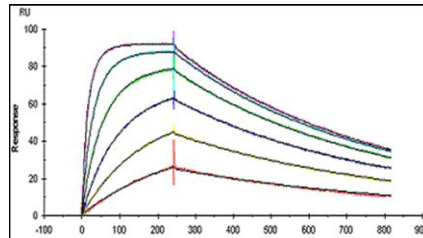


Biotherapeutic Enabling Biology (BioRX) – Lilly Corporate Center (LCC)

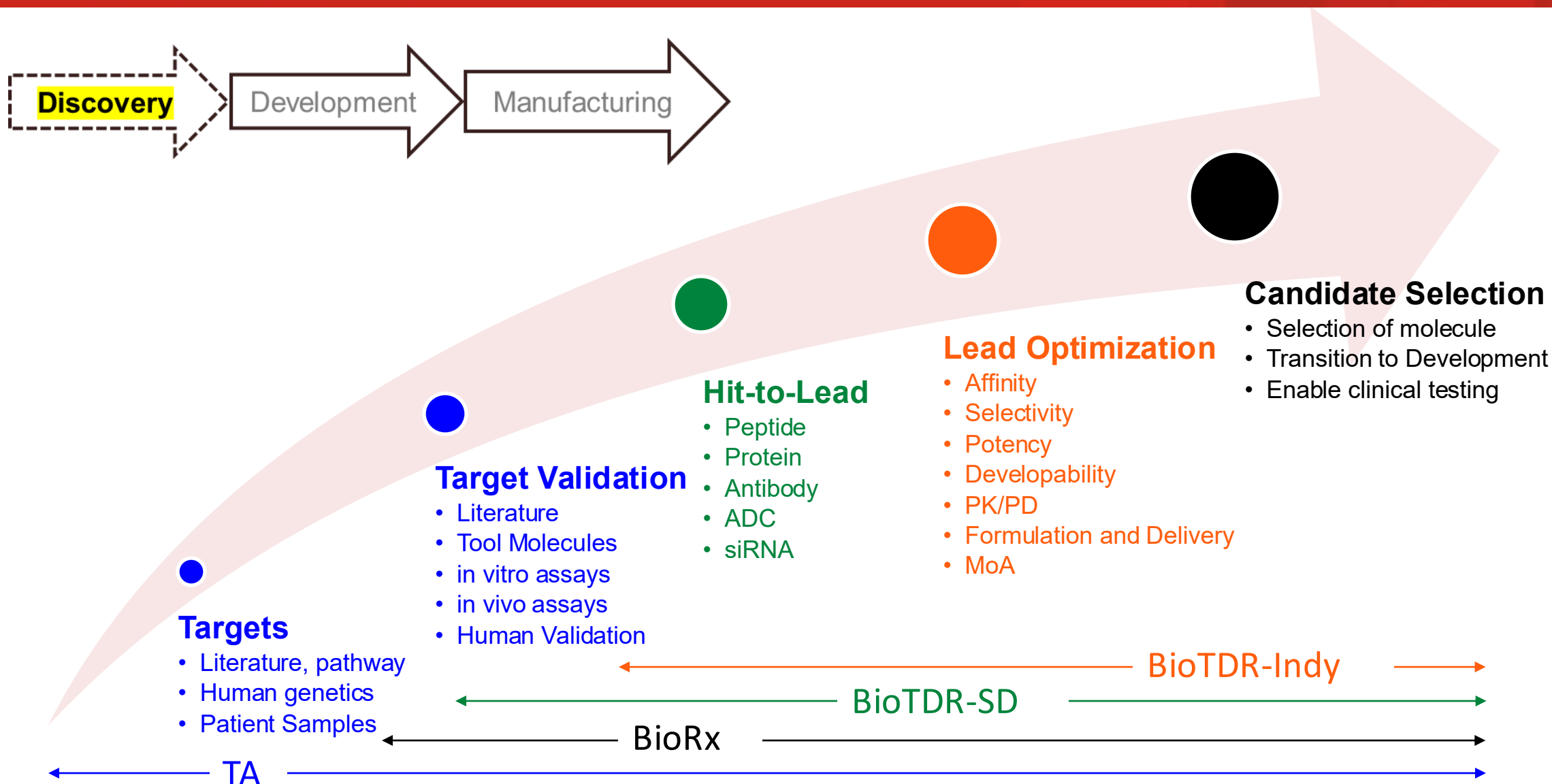
- Deep understanding of disease biology to enable reagent and flow-scheme generation
- *In vitro*, *in vivo*, *ex vivo* assay development

BioTDR Peptide Lead Optimization – LCC and Lilly and Technology Center (LTC)

- Peptide and Protein Engineering
- Expression and Purification
- Protein Characterization
- Formulations & Delivery

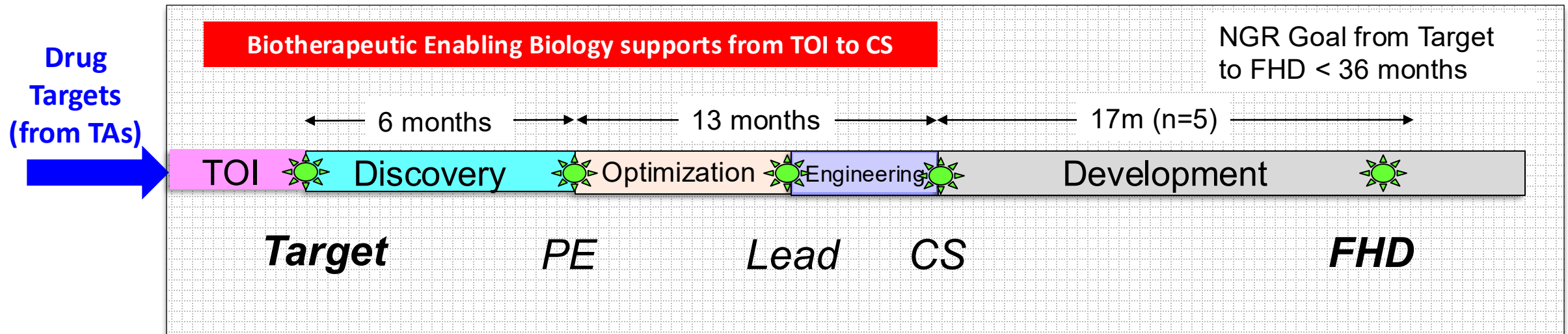


Large Molecule Discovery Process



Biotherapeutic Enabling Biology (BioRX)

Biotherapeutic Enabling Biology - Our mission is to accelerate discovery and delivery of biotherapeutics from TOI to CS. We will root our work on NGR principles and on strong partnerships with BioTDR, TAs and ATP Scientists.



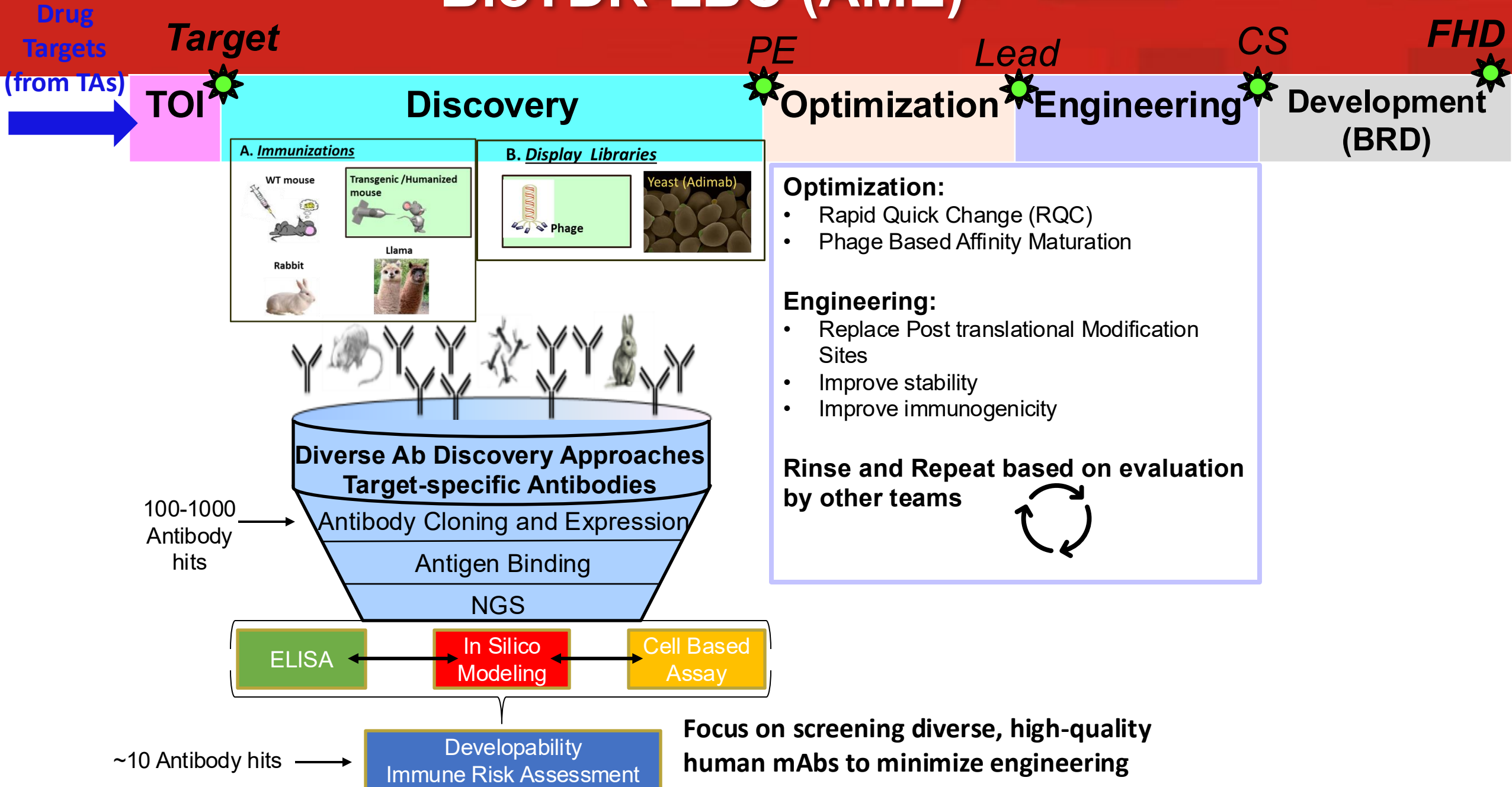
- BioTDR Axis: PO, AME and BioRx work on same TOIs
- TA interface: Target validation
- Develop Ab screening assay before immunization

- Transfer assay to AME for automation as fit
- Functional screen
- Single Cell Analytics for receptor binding and internalization

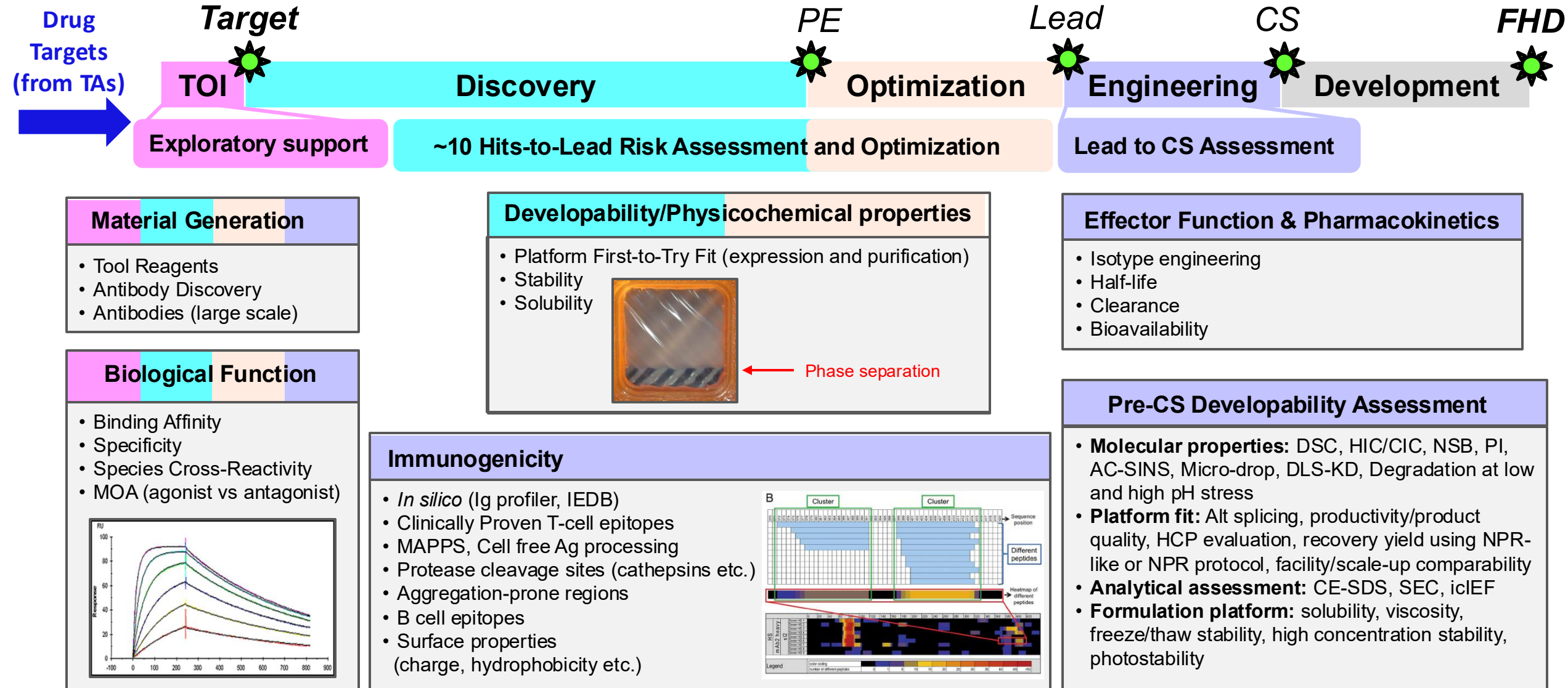
- Innovative assays for 2nd screenings of desired biology (including 1ry cells, Treg, NK, pDCs, ECs, microglia)
- PD models

- Interface with CMC and BR&D
- Effector function assays
- Lot Release Bioassays

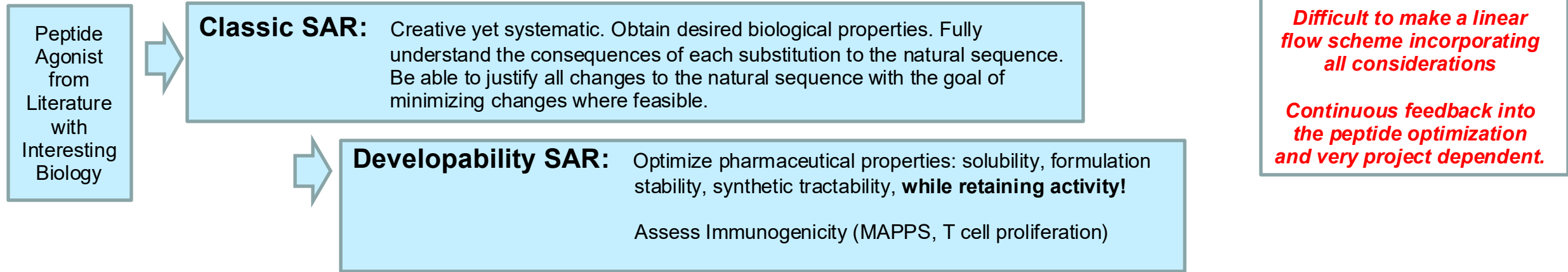
BioTDR-LBC (AME)



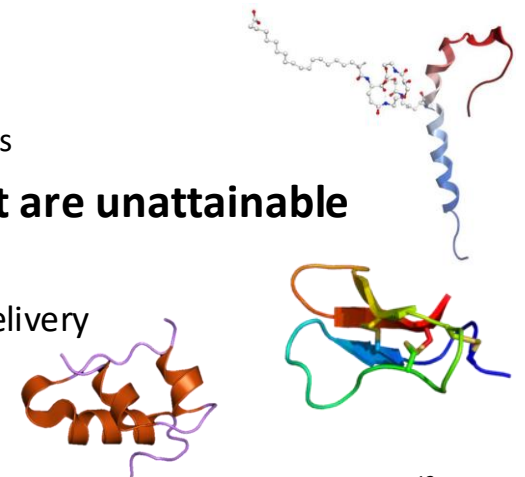
BioTDR-Lead Optimization – Antibodies



BioTech Peptide Group Research Strategy



- **Work with TA's to identify naturally occurring peptide agonists and antagonists, and optimize them for activity, stability, in vivo half-life, and maximize chances of success with "developability"**
 - Larger focus on *agonist* peptides
 - Focus on projects where significant barriers exists for both small molecule and protein / antibody / recombinant approaches
- **Utilize synthetic expertise to interrogate peptide structure-activity relationships in ways that are unattainable and/or complementary to recombinant methods**
 - Example: incorporation of non-coded amino acids into incretin peptides to stabilize against GI proteases for oral delivery
 - Example: insulin SAR for GSI: rapid generation of synthetic insulin analogs



Introduction of Oral Peptide Delivery (OPD) Platform

Oral Peptide Delivery Platform is the combination of cross-functional capabilities and expertise that enable the discovery & development of high-value, orally-bioavailable, peptide-based medicines in a robust, cost-efficient and sustainable manner for multiple therapeutic indications



Scope: Oral Peptide Delivery Platform is focused on **incretins** to address Lilly Diabetes critical business needs



Vision: Lilly is the leader in Oral Incretins by 2030

- ❑ Two oral incretins launches
- ❑ Mature OPD Platform capable of delivering breakthrough medicines



Overall Strategy for Oral Peptide Delivery

Engineering Oral Peptide

Design Principles

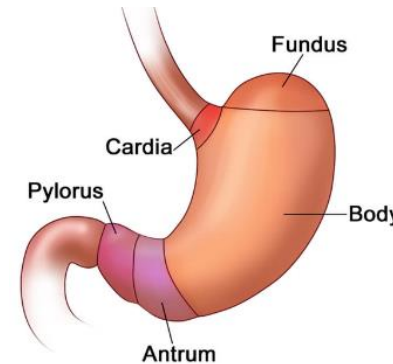
- High Potency
- Stability against proteolysis
- Optimized physicochemical properties (solubility, self-association, etc.)
- Low clearance for dose stacking upon daily oral dosing



Molecular Engineering

- ≥ 10 -fold higher *in vitro* potency than TZP
- Stable in small intestinal fluids/enzymes
- Improved stability in stomach enzyme compared to Semaglutide
- High solubility in pH 5-8 (small intestine pH)
- Long half-life (~7 Days) to achieve dose-accumulation upon oral daily administration
- Low immunogenicity risk

Formulation Strategies for Oral Peptide Delivery



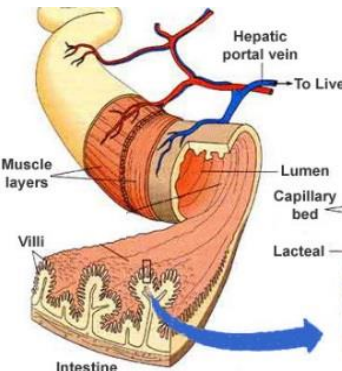
Gastric



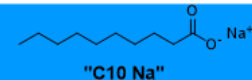
Formulation Strategy:

Erodible and/or IR tablets comprise:

- Peptide (API)
- Gastric permeation Enhancer (PE)

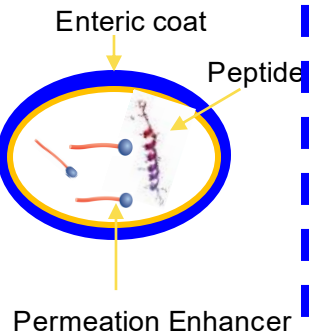


Intestinal



Proximal

Distal

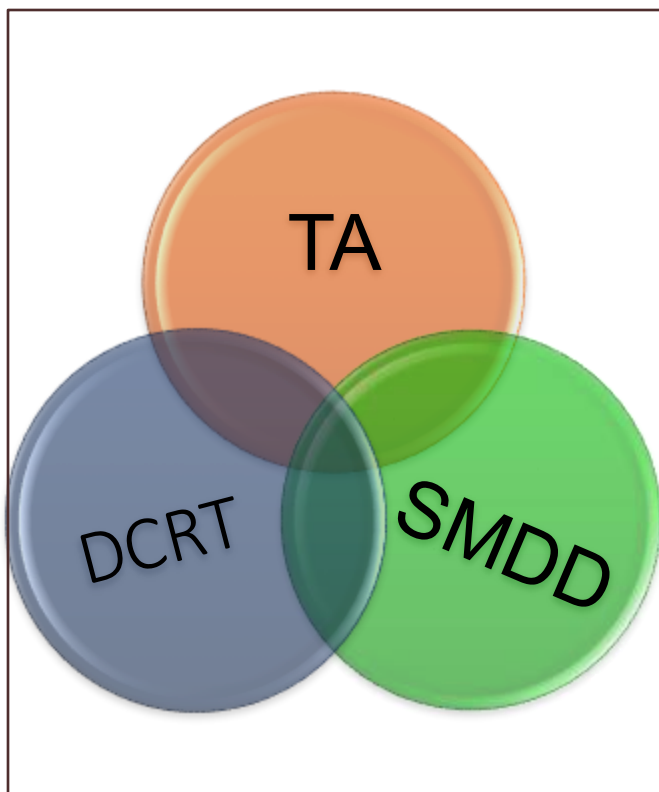


Formulation Strategy:

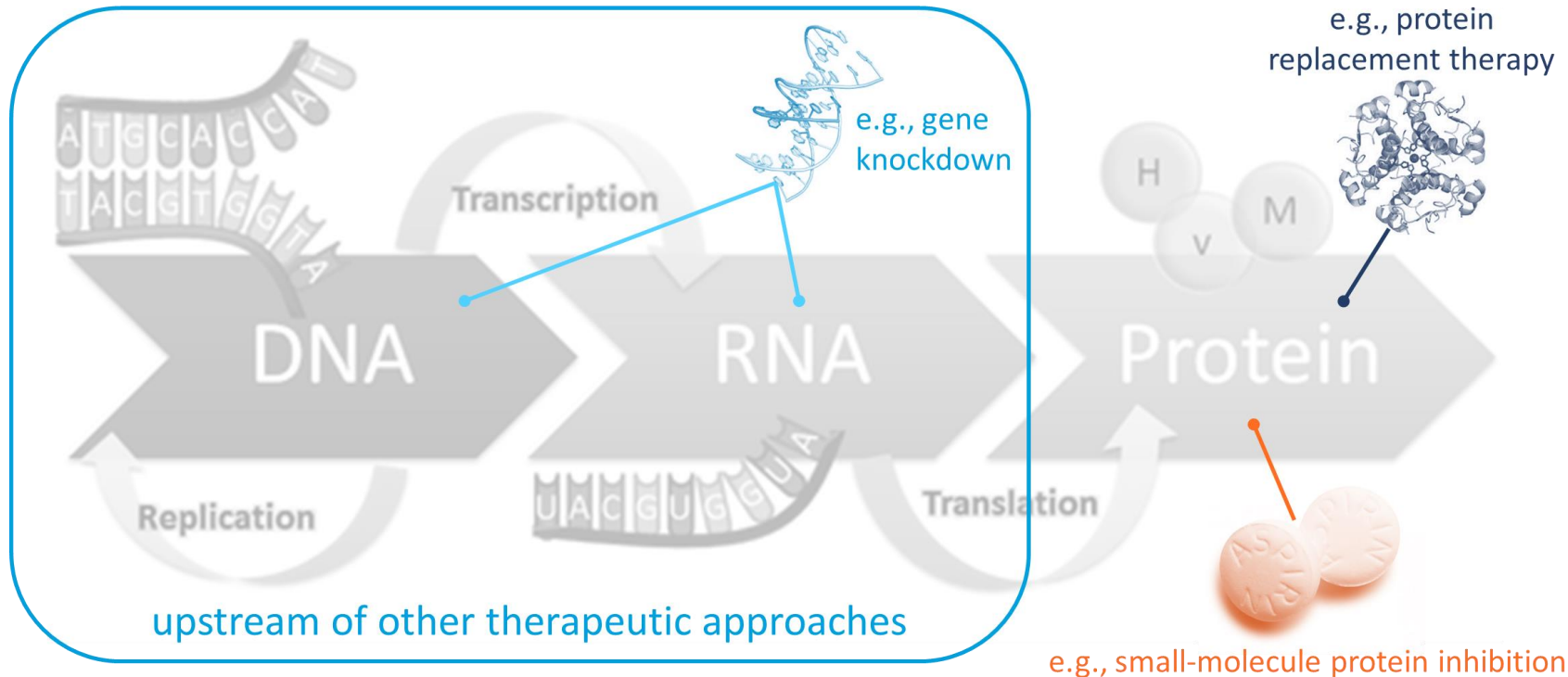
Enteric-coated tablet/capsule comprise:

- Peptide (API)
- Intestinal permeation Enhancer (PE)

Small Molecule Team of Teams



NTM Mission: Establish and expand platforms to access undruggable targets, to deliver breakthrough outcomes for patients



~20,000 human proteins

In 2017 approved drugs only targeted 667 human proteins

Significant opportunity to reach undruggable targets with novel therapeutics

N. Dammes, et al. Trends in Pharmacol Sci.2020

Additional Resources covering "Drug Discovery, Development & Manufacturing"

- **BR&D continuous Learning Series > General > Files**
- **“Bridging the Gap: From Molecule to Medicine” 3-part series by Early Career Professionals in 2019**
 - [Bridging the Gap: From Molecule to Medicine: Part 1](#)
 - [Bridging the Gap: From Molecule to Medicine: Part 2](#)
 - [Bridging the Gap: From Molecule to Medicine: Part 3](#)
- **Lilly-U /LinkedIn/ Learning Plan courses:**
 - [Introduction to the Industry: Section I - Industry Overview](#)
 - [Understanding Pharma: Drug Development Part 1](#)

Lilly Acronyms – always subject to change

TA: Therapeutic Area

BioTDR: Biotechnology Discovery Research

SAR: Structural Activity Relationship

NPR: New Product Resupply; protein expression & purification group within BR&D

PET: Protein Expression Team, group within BioTDR

ATPLV: ADME, Toxicology, Pharmacokinetic and Pharmacodynamic (PK/PD), Laboratory of Experimental Medicine (LEM), Veterinary Resources (V)

ADME: Absorption, Distribution, Metabolism, Excretion

LEM: Laboratory of Experimental Medicine

DCRT: Discovery Chemistry Research & Technology

CMC: Chemistry Manufacturing Control

FHD: First Human Dose

SAD: Single Ascending Dose

MAD: Multiple Ascending Dose

DDCS: Delivery, Device and Connected solutions

API: Active Pharmaceutical Ingredient; also, a department in MIH.

TSMS: Technology Services & Material Science (a group in the API department)

PD: Product Delivery

BRD or BR&D: Bioproduct Research and Development

CSSP: Clinical Systems and Supply Planning

NTM: New Therapeutic Modalities

SMDD: Synthetic Molecule Design and Development

Immunology: Immunology Therapeutics

DRQS: Discovery Research Quality Systems

LQS: Lilly Quality System

PR&D or PRD: Product Research & Development

CI pool: Candidate Intermediate pool

BRIDGE Committee Members



Join BRIDGE on MS teams

Please contact BRIDGE members if you would like to get involved

Big thanks to Management for continued support

Thanks Jirong, Ryan, Ling, Henry, John & Wolfgang

Lilly Biotechnology Center

- Marina Alvi
- Maya Karta
- Jordan Crampton
- Kira Rubtsova
- Sepideh Afshar
- Robb Pagarigan
- Diana Ruiz
- Kevin Separa

Lilly Corporate Center

- Shannon Harlan
- Henning Stockmann
- Angela Okragly
- Jonathan Sims
- Seamus Brennan
- Stephanie Beasley
- Samantha McElyea

Lilly Technology Center

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- Kristen Newburn
- Aya Ryuzoji
- Chetan Patel
- Wei Wang

Please set up a 1:1 or send comments via email to anyone listed above

OR

Drop an anonymous note to BRIDGE **Virtual Drop Box** or interoffice mail **DC0730**